Worksheet # 3 Part A - Multiple Choice
Physics 1I

Name: Key Date: _____

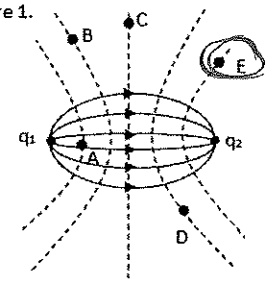
Multiple Choice Questions 1 though 10

1) Figure 1 shows two particles with a net charge, the solid lines are the electric field, and the doted lines are the equal potential lines, if an object with a net negative charge is placed at one of the points indicated by letters A through E, at which point will the object have the largest potential energy?

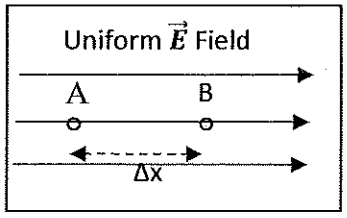
A) A
B) B
C) C
D) D
E) E
F) Cannot say, not enough information

Point E is the highest PE for negative charge

Figure 1.



2) A charged sphere is in a uniform electric field $|\vec{E}|$ and position A is at $V_A = 100$ V and position B is at $V_B = 20$ V, calculate the work that must be done on a positively charged sphere (m = 0.02 kg) with a net charge of 200 μC to move the sphere from Point A to B over a distance of $\Delta x = 0.3$ m.



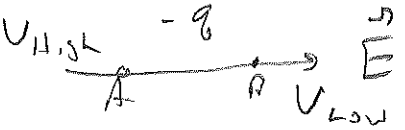
$W = -\Delta PE = -q\Delta V$

$\Delta V = 100V - 20V = 80V$

$W = -(200 \times 10^{-6} C)(80V) = 0.016J$

3) For an electron moving in the direction of the electric field → higher PE

A) its potential energy increases and the electric potential increases.
B) its potential energy decreases and the electric potential increases.
C) its potential energy increases and the electric potential decreases.
D) its potential energy decreases and the electric potential decreases.
E) both its potential energy and the electric potential remain constant.

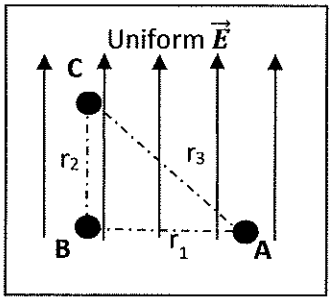


4) Three points in space (A, B and C) are shown in a uniform electric field, if the magnitude of the electric field is $|\vec{E}|$ = 1,000 N/C and the points are separated by $r_1 = 0.3$ m, $r_2 = 0.4$ m and $r_3 = 0.5$ m, what is the potential difference between points A to C?

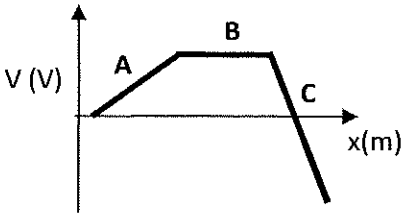
$\Delta V = |\vec{E}|\Delta x = |\vec{E}|r_2$

$\Delta V = 400V$

along $\vec{E}$-field lines

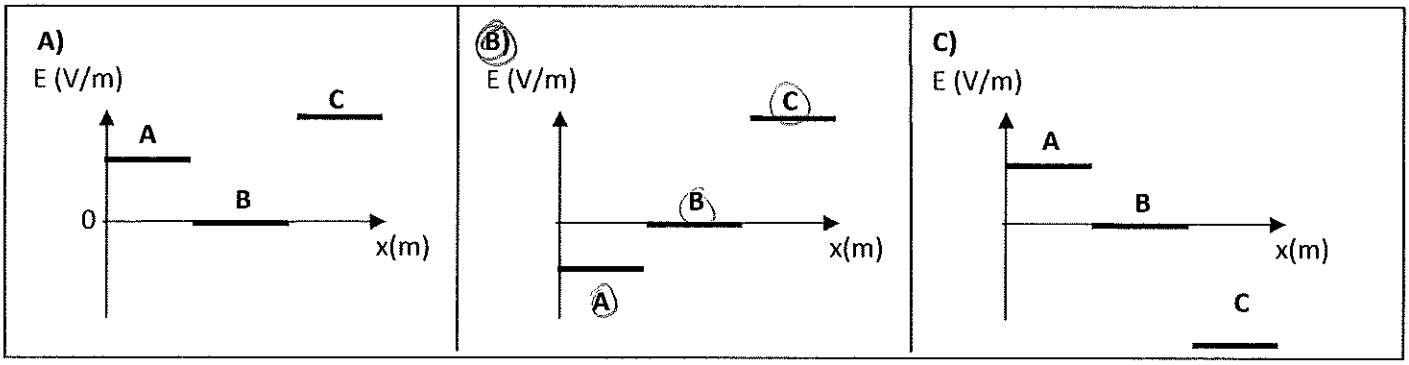


5) For the graph of electric potential vs position, which graph of electric field vs. position best describes the electric field shown in the potential vs position graph?



$\vec{E} = -\left(\frac{dV}{dx}\right)\hat{x}$

$|\vec{E}| = -\frac{dV}{dx}$



6) A sphere has a mass of 2 grams and a net positive charge of 50 μC is accelerated through a potential difference of 10,000 V. Assuming the sphere starts from rest, determine the velocity of the sphere once it leaves the potential?

$m = 2\times10^{-3}$ kg

$q = 50\times10^{-6}$ C

$\Delta V = 10{,}000$ V

$\Delta KE + \Delta PE = 0$

$\Delta KE = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2$

$\Delta PE = -q\Delta V$

$\frac{1}{2}mv_f^2 + (-q\Delta V) = 0$

Solve for v_f

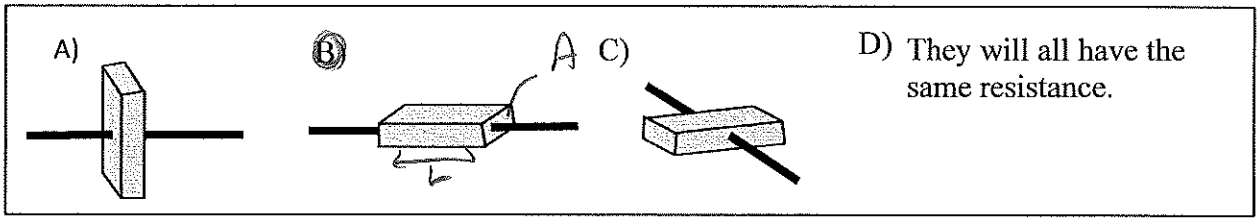
$v_f = \sqrt{\frac{2q\Delta V}{m}} = 22.4$ m/s

7) Which of the following will increase the capacitance of a parallel plate capacitor?

A) Insert a dielectric material between the plates.
B) Increase the potential across the plates.
C) Increase the charge on the plates.
D) All of the above.
E) None of the above.

$C = \kappa C_0$

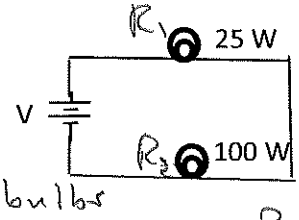
8) A copper block (1cm x 4 cm x 10 cm) is placed in a circuit with three different configurations as shown below, which configuration creates the **largest** resistance in the circuit?



D) They will all have the same resistance.

$R = \frac{\rho L}{A}$

9) A 100 W lamp glows brighter than a 25 W lamp wire in series. The electrical resistance of the 100 W lamp is



A) The 100 W bulb would have ¼ times the resistance as the 25 W bulb.
B) The 100 W bulb would have ½ times the resistance as the 25 W bulb.
C) The 100 W bulb would have 2 times the resistance as the 25 W bulb.
D) The 100 W bulb would have 4 times the resistance as the 25 W bulb.
E) The 100 W bulb would have the same resistance as the 25 W bulb.

Current is same through both bulbs

$P_1 = I^2 R_1 \quad P_2 = I^2 R_2 \quad \frac{\text{Solve for } I^2}{} \quad \frac{P_2}{R_2} = \frac{P_1}{R_1} \quad R_2 = \left(\frac{P_2}{P_1}\right) R_1 = 4R_1$

10) Bob has a light bulb hooked up to a 12 V d.c. power source. In order for Bob to hook up the same light bulb to an a.c. power source and have the light the exact same brightness, what **peak voltage** does Bob need to generate in his a.c. power source?

$V_{d.c.} = V_{rms} \quad \left| \quad V_{rms} = \frac{V_p}{\sqrt{2}} \quad \right| \quad V_p = \sqrt{2}(V_{rms}) = 17V$